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Research Article

Analytical Study on Wireless Sensor Network Communication and Actuators Collaboration

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Wireless sensor networks (WSNs) are used in environmental monitoring, urban planning, healthcare, and industrial automation. These networks must transmit data to succeed. Energy usage, throughput, and latency in the shared wireless medium depend on the medium access control (MAC) layer resource distribution. Data transmission inside networks is vital to their efficiency and efficacy. An analytical research assesses the MAC protocols under consideration and compares their communication properties to WSN routing methods. This paper presents an analytical study of MAC and modeling development of a distributed estimator that addresses the challenges provided by noisy sensory measurements and loss rates in wireless communications. These protocols' suitability for WSN applications is an important goal. Data transfer from sensor nodes to a central sink or gateway requires WSN routing. Proactive, reactive, and hybrid routing approaches are compared for packet delivery ratio, end-to-end latency, communication, and energy economy. Routing protocols are affected by dynamic network characteristics such as node failures and movement in this research.

Keywords: automation; communication; industrial; MAC; routing; sensor; wireless

1. Introduction

The medium access control (MAC) protocols may be classified into three main classes depending on the technique they use for medium access: schedule-based, contention-based, and hybrid-based access mechanisms. Our methodology differs from earlier studies that categorize MAC protocols in wireless sensor networks (WSNs) based on unique energy-saving approaches [1]. Instead, we align with the traditional way of classifying multiple access protocols, as established in earlier studies. The process of communication within WSNs is often organized into many tiers [2]. The primary aim of this methodology is to reduce the likelihood of a collision, and it has demonstrated efficacy in attaining this result. One of the primary aims of the majority of MAC protocols is to ensure that consumers are able to achieve high data transfer rates and experience superior quality of service (QoS). MAC protocols help to enhance energy consumption and lifetime enhancement [3]. In [4] authors proposed the MAC protocol ERAPRAD to define accessing failures of the vehicles. WSNs are a part of the backbone for several applications, including greenhouse farming [5].

This phenomenon is commonly observed in the majority of MAC protocols. In contrast, the wireless sensor MAC protocol prioritizes energy conservation over adherence to QoS standards. The rationale behind this is that the consumption of energy represents a substantial proportion of the costs associated with wireless networks. The energy dissipation experienced during transmissions can be attributed to the utilization of conventional protocols at the MAC layer. The main factors contributing to the dissipation of energy in this context are passive listening, collisions, protocol overhead, and unintended reception.

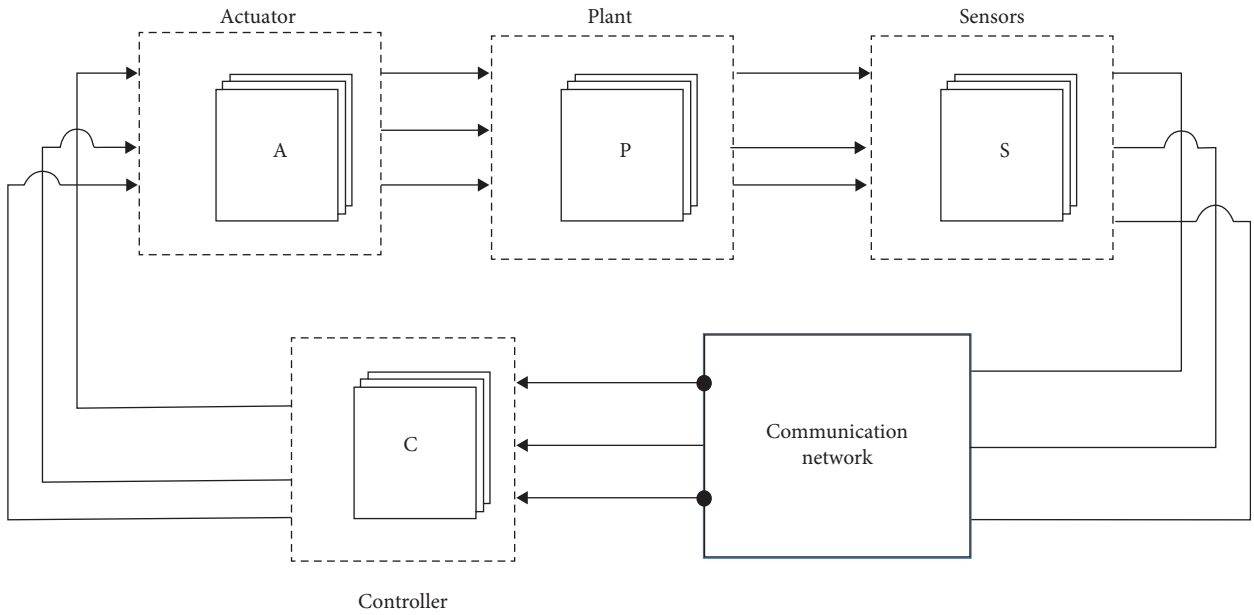

Figure 1: Overview of the WSNs communication and control system.

2. MAC Protocol Applications and Challenges

Numerous MAC protocols have been developed expressly for use in WSNs. The S-MAC, T-MAC, and D-MAC protocols, among others, provide archetypal examples. Sensor networks use MAC protocols that employ an active/sleep strategy to optimize battery life. The MAC protocol effectively mitigates collisions by implementing measures that prohibit simultaneous transmission by two or more interfering nodes. This is done in an effort to reduce the likelihood of a collision occurring. In [6], proposed MAC protocols with LoRaWAN to enhance the wireless communications and address the challenges in different ranges since LoRaWAN can be used over very large distances.

The provision of high throughput as well as excellent QoS is the primary objective of the design process for the majority of MAC protocols. On the other hand, the wireless sensor MAC protocol assigns minimizing energy usage a greater priority than meeting QoS standards [7]. Because of idle listening, collisions, protocol overhead, and overhearing, typical MAC layer protocols are responsible for the loss of a significant amount of energy. The work centers around the IEEE 802.15.4 protocol, which has gained significant popularity as a standard for low-rate and low-power WSNs across different domains of application. In [8], sensing protocol of MAC (DCSM) was proposed, which helped to reach each node and control the traffic changes and slot.

To characterize the basic limitations of this protocol and optimize its settings, it is crucial to comprehend dependability, latency, and throughput. However, a comprehensive comprehension of the attainable performance of this hybrid MAC has not been established at present. The primary obstacle to achieving precise analysis lies in the simultaneous presence of stochastic elements in the random access process and deterministic elements in the time division multiple access (TDMA) scheme. Markov chains are employed for the purpose of modeling the contention access scheme and the behavior of the TDMA access scheme within the IEEE 802.15.4 protocol. These models have been validated through theoretical evaluation and Monte Carlo models. The evaluation of network performance, including reliability, average packet delay, average queueing delay, and throughput, is conducted through the utilization of this novel model. Additionally, it has been demonstrated that the performance of the hybrid MAC exhibits notable deviations from the findings previously documented in scholarly literature. The MAC protocols have been proposed to enhance the various applications, including communications, algorithms, mathematical problems, including fuzzy logic methods optimization [9].

Moreover, it can be inferred that the balance between the throughput of the random access and the TDMA scheme plays a crucial role in maximizing the overall throughput of the hybrid MAC.

In the command and control architecture of integrated closed-loop networks over a wireless network, there are N controllers that are responsible for monitoring and controlling N plants. As illustrated in Figure 1, it involves the management of multiple plants through a wireless network. The sensors periodically or sporadically sample the outputs of the plants, which are then transmitted to the controller via a network. Upon receiving the measurements, the controller proceeds to compute a new control command. The control signal is transmitted to the actuator that is connected to the plant. The IEEE 802.15.4 standard does not need to be altered in any way for the method to operate, and it is simple enough to be implemented on any network node. On a test bed comprised of commercially available wireless sensor nodes, an experimental implementation of the protocol is carried out and assessed. The analysis is correct, the suggested method satisfies the reliability and latency criteria, and the technique assures a longer lifespan for the

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infrastructure under both stable and transient network circumstances. Based on hybrid range and angle measurements, the target location is determined, and data transmission security issues are mitigated [10].

3. MAC Protocols and Routing Discussion and Analysis

These proposals have originated from both academic sources (e.g., [11, 12]) as well as business sources (e.g., [13]). This section focuses on the protocols that have emerged in recent years, specifically those that are relevant to the categories of applications under investigation in this study. In recent years, numerous protocols have been suggested for WSNs to cater to diverse applications, including area monitoring, monitoring the environment, and networks.

Table 1 shows the studied protocols and routing, and the table keys are: P, energy; L, reliability; and C, communication delay. * protocol was created taking the column indication into account, + indicates that the technique was created with the indication in mind and was empirically verified, − indicates that although the outcomes of a simulation or experiment contain an indicator, the protocol design does not, and hence cannot, regulate it. ✓, scheduling; O, hybrid; Ø, contention; D, data-centric. #, MAC; R, routing; T, topology; and 0, location.

To avoid collisions, MAC layers prohibit several interfering nodes from broadcasting simultaneously. This reduces data loss risk. This will enable network efficiency. Most MAC protocols aim to provide fast throughput and great service. Many MAC protocols have been developed for WSNs. S-MAC, T-MAC, and D-MAC are examples of common scenario kinds. Sensor networks use MAC protocols with an active/sleep strategy to save battery life. MAC protocol prevents collisions. The above approach mitigates collisions by preventing simultaneous broadcasting by two interfering nodes. Wireless networks lose packets and experience delays. Thus, the network may affect closed-loop stability, as shown in Figure 1. Provides an adaptive 802.15.4 of the IEEE protocol that changes CSMA/CA algorithm MAC settings to provide reliable, timely, and energy-efficient communications. We recommend utilizing a generalized Markov chain to depict these linkages using simple expressions without losing model fidelity. After that, the model is utilized to build an adaptive method to minimize power consumption while maintaining reliability and packet-level transmission delay restrictions.

In Figure 2, MAC protocols and routing function analysis is shown, the analysis has been taken for different side, such as account indication of the created protocol (*), created techniques (+) shows that the method was developed with the indicated of effectiveness and was confirmed by empirical evidence, and (−) shows that even when a simulation or experiment's results, include an indication, the protocol layout doesn't and so lacks the capacity to control it, which all these shows the effectiveness in the delay communications, reliability, and energy. Figure 3 shows the (a) performance percentages of examined MAC protocols and mechanisms and (b) percentage of the selected MAC protocols and routing in this study.

Access schemes within the MAC protocols and routing models have been validated through theoretical evaluation and Monte Carlo models. The evaluation of network performance, including reliability, average packet delay, average queueing delay, and throughput, is conducted through the utilization of the model. Figure 4 shows the access scheme analysis for MAC and routing. Additionally, it has been demonstrated that the performance of the hybrid MAC exhibits notable deviations from the findings previously documented in scholarly literature. Moreover, it can be inferred that the balance between the throughput of the random access and the TDMA scheme plays a crucial role in maximizing the overall throughput of the hybrid MAC.

4. Wireless Sensor and Actuator Networks (WSANs) Control System Modeling

The WSANs applied to industrial instrumentation and control applications, in our opinion, are very functional for use in the NCS [14]. Consider an HVAC system, for instance. Room temperatures are considered state variables in our opinion. Some subjective "set points" must be observed in order to avoid discomfort. The goal is to be able to configure all system variables to fulfill predetermined goals. Instead of only using a control mechanism integrated inside the actuators, we concentrate on the DC approach. These models provide a base from which more advanced systems may be built. Plant dynamics are observed and characterized by plant attributes, which may be expressed in spatial time as follows:

$$y[k] = \beta[k] + \mathrm{Cx}[k-1] + \mathrm{Du}^{a}[k], \tag{1}$$

where D and C are the suitable dimension (constant-matrices). β is the noise mean of zero. In our opinion, we believe each xj is tracked by a separate sensor2, where 1 1, …, $n^{j}a$ is $j\ of\ Q_j$ and Q stands for the sensor and L stands for the actuator.

A subdomain of sensors may adopt specific data aggregation techniques to give control units more accurate data, as is seen in industrial applications. For the purpose of measuring the temperature, sensors are placed in place to report on the actuators (air conditioners, for example).

This is due to the fact that the actuator output may influence the temperature (and all other operating parameters, such as noise, ventilation, etc.). The equation is considered to be linear in time (1/1). Since, by assumption, we believe that the present room temperatures are proportional to their prior values and the actuator output, we may safely conclude that. The physical process graph, that is, $G = \Delta(U, E)$. The collection of vertices in a directed graph is $\mathrm{Um} = \{\mathrm{Li}\} \cup \{\mathrm{Qj}\}$ and the set of edges is $\mathrm{Em}.\forall u \in \mathrm{Um}$, let the variable that corresponds to u be $y(\text{if } u \in \{Q\})$ or $v_a(\text{if } u \in \{\mathrm{Li}\})u$, edge to uJ $(\mathrm{uJ} \in \mathrm{Um})$, that is, $\exists \mathrm{Em}(u, \mathrm{uJ})$, if the variable corresponding of u^{j} is to a degree affected by being evaluated by either Equation (1) or Equation (4)

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Table 1: The selected MAC protocols and routing analysis.

MAC protocols and routing	Class	Access scheme	Analysis	P	L	C
SMACS [16, 17]	#	✓		*		
PICORADIO [18]	#	✓		+	−	
PACT [19]	#	✓		*		
HyMAC [20]	#	X	C	*	−	−
EAT [21]	#	✓		*	*	−
TRAMA [22, 22, 23]	#	✓		*	−	−
LMAC [15]	#	✓		*		
MAMAC [24]	#	✓	C	*	−	*
Z-MAC [25]	#	X	C	−	*	−
CSMA/TDMA [19, 26, 27]	#	Random	P, L, C	*	*	*
EEMNS [28]	#	✓		+	−	+
PEDAMACS [29]	#	✓		*		*
MMSN [30]	#	✓		+	*	−
CFFD [31]	#	✓		−	+	−
L-CSMA [32]	#	Ø	C			
DOZER [33]	#	✓		+	+	
PMMAC [34]	#	✓			+	
TSMP [35, 36]	#	✓		+	+	−
SPAN [37, 38]	#	Ø	C	*	−	−
LPL [39, 40]	#	Ø		+		
STEM [41]	#	Ø	P, C	*		*
LEACH [42, 43]	R	D	p	*	−	−
CYCLED RECEIVER [44, 45]	#	Ø	P	*		
WISEMAC [46]	#	Ø	P, C	+		+
BREATH [47]	R	0	P, L, C	+	+	+
KOALA [48, 49]	#	Ø		+		
SMAC [50, 51]	#	X	C	+	−	−
TMAC [52]	#	X		+		
ORCHESTRA/RMS in ETS [53]	R	0	P, L, C	+	+	+
DSMAC [54]	#	X	C	*		*
DMAC [55]	#	X		*	−	*
IRIS [56]	#	X		+	+	
SCP [57, 58]	#	X	P	+	−	−
CRANKSHAFT [59]	#	X		+	−	−
Z-MAC [60]	#	X	L	+	+	−
RPL [61, 62, 63, 64]	R	T		+	+	+
CTP [65, 66]	R	T		+	+	
BCP [67]	R	T	P	−	+	+
DCF [68]	#	✓		+	+	+
REACT [69]	R	D		*	*	
AEA FCB [70]	R	D	P	*	—	
REEP [71]	R	T	P	−	+	+
DIRECTED DIFFUSION [72]	R	D	P	*	−	+
ELASTIC [73]	#	X	C	*	—	*

the variable connected to it of ν. Two sets to define: a sensor affected by the effects of $L_i \, \forall i \in \{1 \ldots n_\alpha\}$

$$Q^{\mathrm{Li}} = \left\{ Q_j \middle| Q_j \in U_p, \exists E_m\left(L_i, Q_j\right) \right\} = \left\{ Q_j \middle| d_{j,i} \neq 0 \right\}, \tag{2}$$

the taking actuators:

$$\begin{aligned} &Q_j \, \forall j \in \{1, \ldots, n_s\} \\ &L^{\mathrm{Qj}} = \left\{ L_i | L_i \in U_m, \exists E_m\left(L_i, Q_j\right) \right\} = \left\{ L_i | d_{j,i} \neq 0 \right\}. \end{aligned} \tag{3}$$

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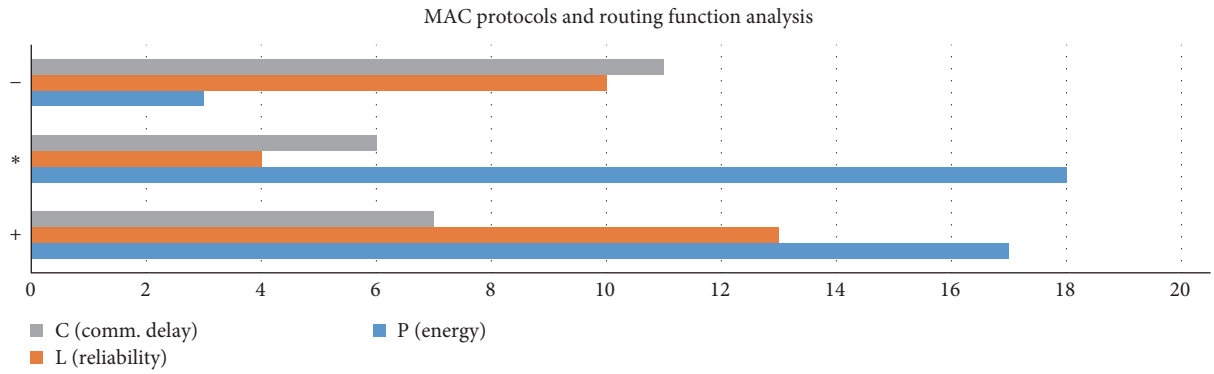

Figure 2: MAC protocols and routing function analysis.

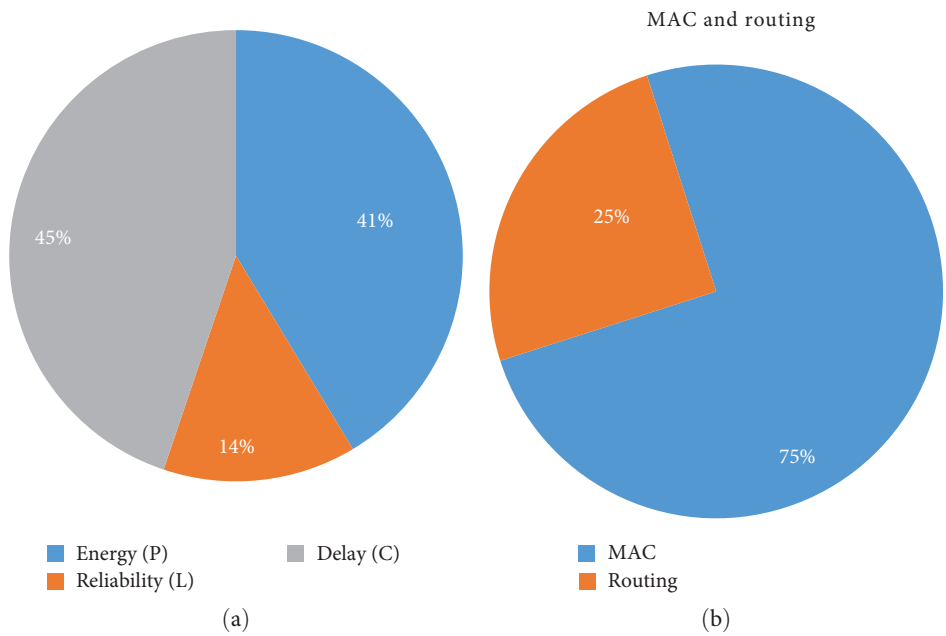

Figure 3: (a) Performance percentages of examined MAC protocols and mechanisms. (b) Percentage of the selected MAC and routing over this study.

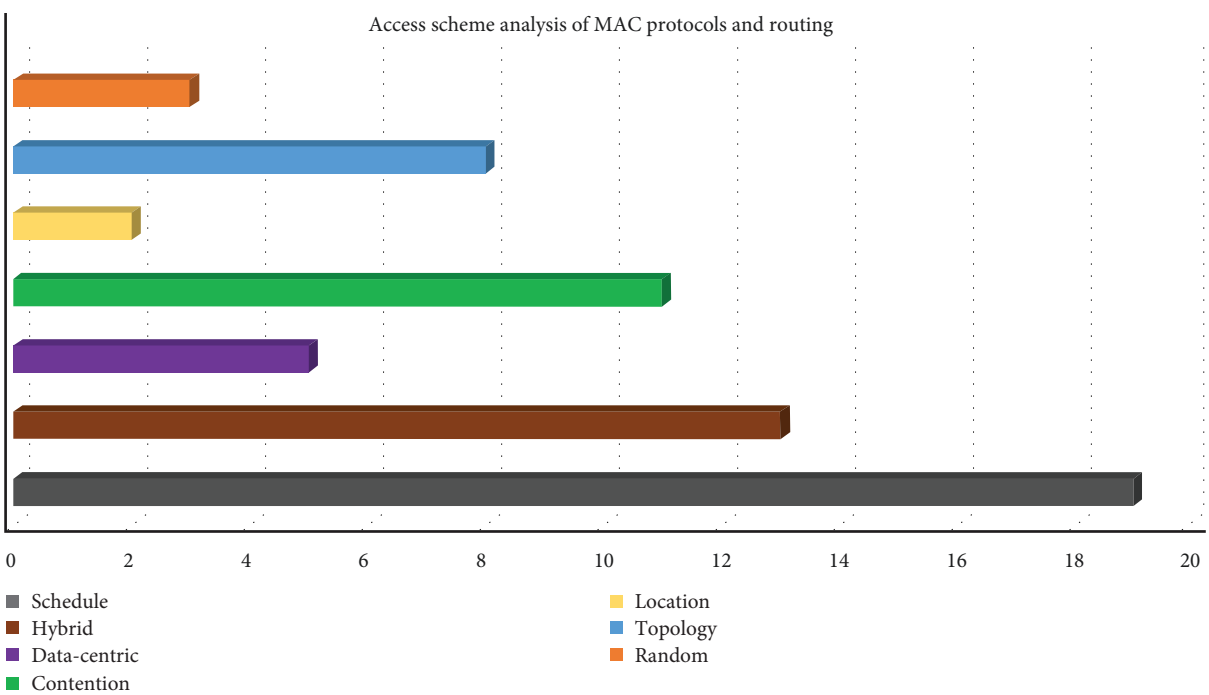

Figure 4: Access scheme analysis of MAC and routing.

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Then

$$\begin{aligned} y_j[k] &= C_{j,j}y_j[k-1] + \sum_{\mathrm{Li}\in L^{\mathrm{Qj}}} d_{j,i}v_i^{\alpha} + \beta_j[k] \\ &\quad + \sum_{1\le \iota\le n_{Q\iota}\neq i} c_{j,i}y_i[k-1] + \sum_{\mathrm{Li}\in L^{Q_j}} d_{j,i}v_i^{\alpha}[k] \\ &= C_{j,j}y_j[k-1] + \beta_j[k], \end{aligned} \tag{4}$$

α_i is the noise, the noise that represents the extraction of a true output from that predicted by $Q_i - \alpha_i$.

The graph communications networked, that is, $G(\mathrm{Uc}, \mathrm{Ec})$, which is an isotonic solution with $\mathrm{Uc} = \mathrm{Um}$. u and uj having an edge $(u, \mathrm{uj} \in \mathrm{Uc})$, that is, $\mathrm{Ec}(u, \mathrm{uj})$ if they are linked by wireless connections. assume that network is linked in such a way that any sensor may communicate with any of its associated actuators and vise versa.

The sensor Q_j gave effect of $y_j[k]$ is supposed to be equal:

$$x_j[k] = y_j[k] + \gamma_j[k]\}, \tag{5}$$

and

$$m \triangleq |L^{\mathrm{Qj}}|, j = 1, \ldots, n_s. \tag{6}$$

From $v^a[k-1]$ to $v^a[k]$,

$$v^a[k] = v^a[k-1] + v_i[k] + \alpha_i[k], \tag{7}$$

then

$$f[k] \triangleq \frac{1}{n_Q}\sum_{i=1}^{n^Q}(m_i - y[k])^2. \tag{8}$$

As shown in Figure 5, we present collaboration control distribution (CCD) approach that combines, distribute estimate optimum control and locally collaborations.

5. Algorithm for Distributed Estimation

In this part, we provide a distributed estimation method that is controlled by all actuators to estimate $\{y_j\}$,which the control of a single actuator is reliant on the data that can get from its sensors that are affected by the actuator. The method in which every sensor uses just the information of the sensors next to it, sometimes with the addition of the Kalman filter, is known as a consensus-based distributed estimating technique. A major limitation of these processes is that they rely on sensor networks where the sensors get the same process state. In this article, two reasons for the problems estimation:

1. A variety of physical factors are measured by the sensors that is $\{y_j\}$.
2. Each actuator does estimate based on process state observations received from the sensors.

In the following, we make no assumptions about packet delay and solely consider packet loss. Furthermore, provided a discrete step time, we may count the packets with sufficiently long delays. In this scenario, the estimates under packet delay may be evaluated in the same way as packet loss can.

$$\widetilde{y}_j^{(i)}[k] \triangleq E\{y[k]|\widetilde{X}[k]L_j\}. \tag{9}$$

That means $y_j[k]$ estimate is $\widehat{y}^{(i)}[k]$ in case of L_i, we model channel of the wireless between actuators and sensors. Moreover, Figure 6, shows the possibilities for dynamic relocate, decentralization and adding sensors or actuator without need for shutdown the system.

This means, $\widehat{y}_j^{(i)}[k]$ is Li's estimate of $y_j[k]$. We simulate the communication links between sensors and actuators using the erasure model described in [15]. $\forall i \in \{1\ldots n_Q\}, \forall i$ in such a way that $L_i \in L^{Q_j}$ and $\forall k > 0$, describing the Q_j to L_i packet transfer outcome as:

1. If $x_j[k]$ is received by L_i, $\omega_{ji}[k] = 1$, sensor current measurement, (estimation step) way of linearity, that is:

$$\widetilde{y}_j^{(i)}[k] = [k]\widetilde{y}_j^{(i)}[k-1] + [1 - t[k])x_j[k]. \tag{10}$$

2. Furthermore, historical data may be used in this instance. If we depend on only historical data in this case, the estimate would be too old and even useless.

$$|y_j[k] - m_j| \le \Delta, \text{ where } K < k, \tag{11}$$

$$y_j^{(i)}[k] = [k]\widetilde{y}_j^{(i)}[k-1] + [1 - z[k])\ m_j, \tag{12}$$

with using t as a constant instead of $t[k]$ and $z[k]$ if $|t| < |z|$ and Equation (11) hold

$$\lim_{k\to\infty}\left|\mathrm{E}\left\{e_j^{(i)}[k]\right\}\right| \le \frac{|\omega_{ji}(t-z) + z|}{1 - |\omega_{ji}(t-z) + z|}\Xi_{ji}, \tag{13}$$

where

$$\Xi_{ji} = \left(1 + \left|\frac{\omega_{ji} - \omega_{ji}(t-1)}{\omega_{ji}(t-z) + z}\right|\right)(\Delta + |m_j|) + \left|\frac{1 - \omega_{ji}(1-z)}{\omega_{ji}(t-z) + t}\right|.|m_j|, \tag{14}$$

the proof: algorithm estimation shown in Equations (10, 12) summarized below:

$$\begin{aligned} \widehat{y}^i|K| = \omega_{ji}[k]\Big(t\widehat{y}_{ji}[k-1] + (1-t)X_j[k] + \Big(1 + \omega_{ji}[k]\Big) \\ \Big(z\widehat{y}_{ji}[k-1] + (1-t)m_j\Big). \end{aligned} \tag{15}$$

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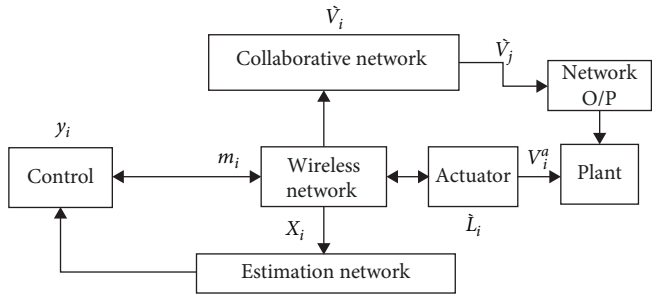

Figure 5: Control collaborative and distribute (CCD) by using actuator components.

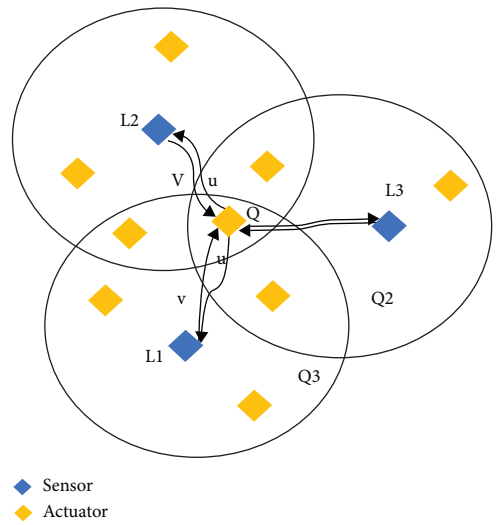

Figure 6: Procedure of control collaborative distribute.

The error of estimation (e) is:

$$\begin{aligned} e_j^{(i)}[k] &= \left(\omega_{j^i}[k]t + z - \omega_{j^i}[k]z\right)e_j^{(i)}[k-1] \\ &+ \left(\omega_{j^i}[k]t + z - \omega_{j^i}[k]t\right)y[k-1] \\ &+ \left(\omega_{j^i}[k] - \omega_{j^i}[k]t - 1\right)y_j[k-1] \\ &+ \left(\omega_{j^i}[k](1-t)\gamma_j[k] + \left(1 - \omega_{j^i}[k]\right)(1-z)m_j, \end{aligned} \tag{16}$$

Then,

$$\begin{aligned} \left|E\left\{e_j^{(i)}[k]\right\}\right| &\le \left|\left(\omega_{j^i}(t-z) + z\right|\left|E\left\{e_j^{(i)}[k-1]\right\} + \Xi_{j^i}\right| \\ &\le \Xi_{j^i}\frac{\left|\omega_{j^i}(t-z) + z\right| - \left|\omega_{j^i}(t-z) + z\right|^{k+1}}{1 - \left|\omega_{j^i}(t-z) + z\right|} \\ &+ \left|\omega_{j^i}(t-z) + z\right|^k\left|E\left\{e_j^{(i)}[0]\right\}\right|, \end{aligned} \tag{17}$$

where Equation (11) and $\gamma[K]$ that has 0 main is used in the aforementioned deduction.

It can be easily seen that:

$0 \le \omega_{j^i} \le 1$; hence $\left|\omega_{j^i}(t-z) + z\right| \le \max\{|t||z|\} < 1$, and Equation (13) can be proven by taking the limit $k \to +\infty$

$$\lim_{k\to\infty} E\left\{\left(e_j^{(i)}[k]\right)^2\right\} \le \frac{\left|\omega_{j^i}(t^2 - z^2) + z^2\right|}{1 - \left|\omega_{j^i}(t^2 - z^2) + t^2\right|}\Xi'_{j^i}. \tag{18}$$

The weight of previous data in the present estimate is specified by the parameter t. When t is set to 0.95, the CCD's performance with regard to different values of t is illustrated in Figure 7. This graph suggests that the appropriate range of t is (0, 0.47). When t is too high, the previous estimate overpowers the present estimate, preventing the sensory measurements from updating their estimate in real-time. However, when t is too small, measurement disturbances will cause the estimate to be inaccurate. The variable t is in charge of instances of packet loss. At a fixed packet loss value, the impact of adding the steady state into the estimation process is seen in Figure 5, as estimates and control at 0 z <1 beat those when $z = 1$. When t is constant, we can also observe that the optimum t is linked to the packet transmission-reception ratio. As rises, the optimum z increases until it equals one if we also use σf as the criterion. the product of actual noise and an additive white Gaussian noise with a frequency range of 0%–30%, as shown in Figure 8.

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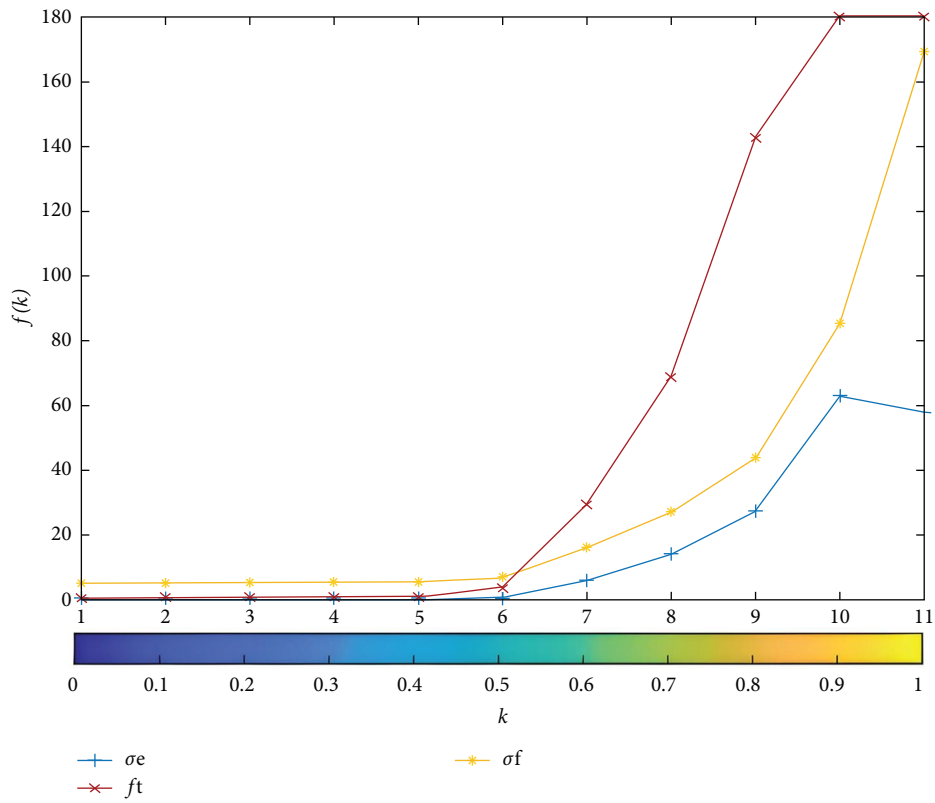
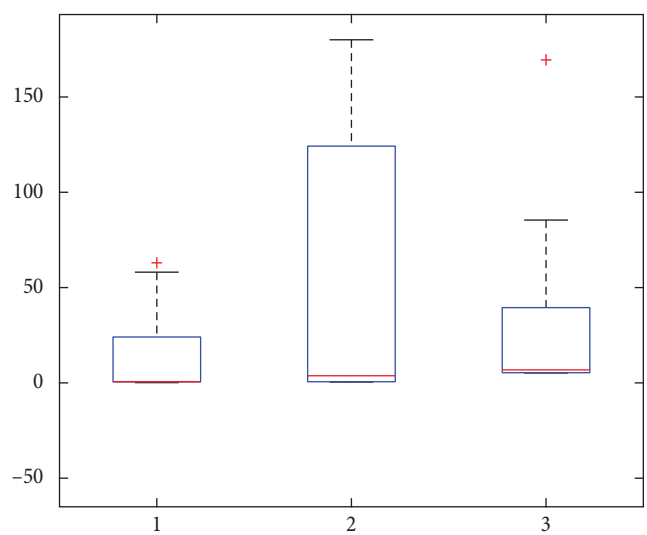


(a)



(b)

Figure 7: (a, b) Performance control with respect to t'Q.

6. Conclusion

The present study conducts an analytical study into the realm of WSNs communications, with a special emphasis on examining MAC protocols and routing. This finding indicates that the choice of a MAC protocol has a substantial impact on the energy efficiency, latency, and throughput of WSNs. TDMA, CSMA, and contention-based MAC protocols each possess distinct benefits and drawbacks. The identification of the most suitable MAC protocol is aided by the examination of parameters, such as network size, traffic patterns, and energy constraints, which have importance for both scholars and professionals in the field. The analysis of routing is a crucial component of communication in WSNs. The scalability of WSNs may be impacted by several aspects, including reliability, dependability, and energy consumption proposed strategy for

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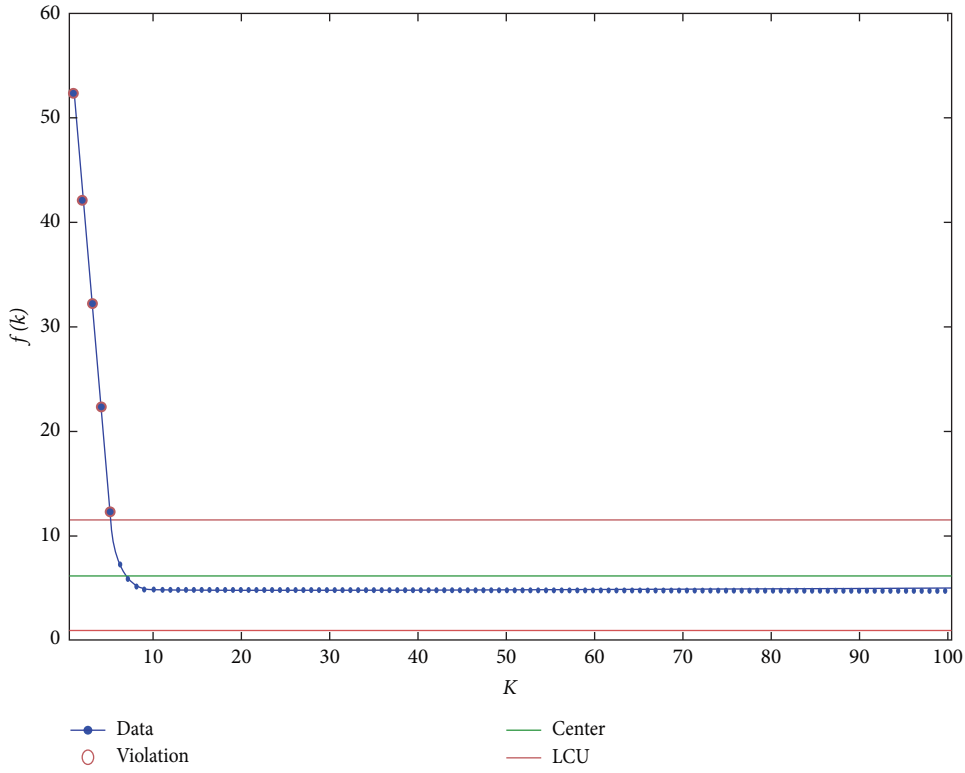
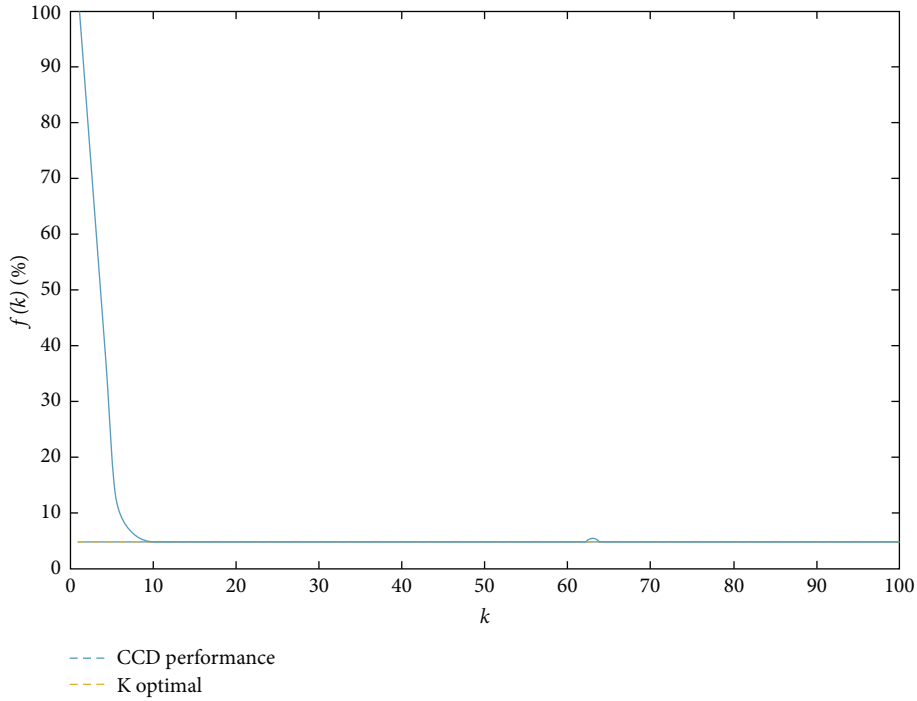


(a)



(b)

Figure 8: CCD Performance "m_{noise} is the power of the system noises" (a) without terms (b) with terms.

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implementing a CCD approach in industrial control systems. The concept utilizes WSANs to address challenges related to sensor measurement errors and packet loss in wireless communication. Additionally, a distributed estimation technique is created to enhance the overall system performance. The aforementioned approach is used to implement a control methodology that facilitates local cooperation and maximizes the utilization of interactions between actuators and sensors. The effectiveness of the approach is shown by comprehensive simulation results.

The alignment of routing techniques with the objectives and constraints of the network is of utmost importance. Empirical research has shown the need for synchronization in WSNs, especially in those using TDMA-based MAC protocols. The process of harmonization enhances the utilization of channels and enhances the efficiency of networks by efficiently addressing and resolving disputes or disagreements. The investigation undertaken in our research elucidated the effects of modifications in MAC protocols and routing methods on WSNs. Future studies in the field of WSNs will include the investigation of energy saving, flexibility, decentralization, and the facilitation of support for diverse applications.

Data Availability Statement

The datasets generated and analyzed during this study are not publicly available but can be available upon reasonable request from the corresponding author.

Conflicts of Interest

The authors declare no conflicts of interest.

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